

MAT205: Abstract Algebra II

Rings, Polynomial Rings, and Polynomial Functions

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Part I

Rings and Ring Homomorphisms

Definition: Ring

Definition. A **ring** R is a set with two binary operations

$$+ : R \times R \rightarrow R, \quad \cdot : R \times R \rightarrow R$$

such that:

1. $(R, +)$ is an abelian group;
2. multiplication is associative;
3. multiplication distributes over addition:

$$a(b + c) = ab + ac, \quad (a + b)c = ac + bc.$$

The additive identity is written 0 .

Example: Basic Rings

Ring	Comment
$\mathbb{Z}$	integers
$\mathbb{Q}, \mathbb{R}, \mathbb{C}$	fields
$\mathbb{Z}_n$	modular arithmetic
$M_n(F)$	matrices over a field
$\mathcal{F}(\mathbb{R}, \mathbb{R})$	real-valued functions
$R_1 \times R_2$	product ring

In $M_n(F)$, multiplication is usually not commutative.

Definition: Commutative Rings and Unity

Commutative ring.

$$ab = ba \quad \text{for all } a, b \in R.$$

Ring with unity. There is an element 1_R such that

$$1_R a = a 1_R = a.$$

For most of this course, our default examples are commutative rings with unity.

Definition: Units

Let R be a ring with unity.

An element $u \in R$ is a **unit** if there exists $v \in R$ such that

$$uv = vu = 1_R.$$

The units form a group:

$$R^\times.$$

Examples:

$$\mathbb{Z}^\times = \{\pm 1\}, \quad \mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}.$$

Definition: Fields

Definition. A **field** is a commutative ring with unity $1 \neq 0$ in which every nonzero element is a unit.

Examples:

$$\mathbb{Q}, \quad \mathbb{R}, \quad \mathbb{C}, \quad \mathbb{F}_p = \mathbb{Z}_p.$$

Non-example:

$$\mathbb{Z}$$

is not a field, because 2 has no inverse in $\mathbb{Z}$.

Definition: Zero Divisors

Definition. A nonzero element $a \in R$ is a **zero divisor** if there exists nonzero $b \in R$ such that

$$ab = 0.$$

Example in $\mathbb{Z}_6$:

$$\bar{2} \cdot \bar{3} = \bar{0}.$$

So $\mathbb{Z}_6$ has zero divisors.

Definition: Integral Domains

A ring has **no zero divisors** if

$$ab = 0 \implies a = 0 \text{ or } b = 0.$$

A commutative ring with unity $1 \neq 0$ and no zero divisors is an **integral domain**.

Examples:

$$\mathbb{Z}, \quad F, \quad F[x], \quad F[x_1, \dots, x_n], \quad F[t, t^{-1}].$$

Example: More Zero Divisors

Product rings usually have zero divisors:

$$(1, 0)(0, 1) = (0, 0) \quad \text{in } R \times S.$$

Matrix rings have zero divisors:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Some rings have nilpotents:

$$\bar{2}^2 = \bar{0} \quad \text{in } \mathbb{Z}_4.$$

Definition: Ring Homomorphisms

Let R, S be rings.

Definition. A map $\varphi : R \rightarrow S$ is a **ring homomorphism** if

$$\varphi(a + b) = \varphi(a) + \varphi(b),$$

and

$$\varphi(ab) = \varphi(a)\varphi(b).$$

Modern convention often also requires

$$\varphi(1_R) = 1_S.$$

Proposition: Point Evaluation of Functions

Let

$$\mathcal{F}(\mathbb{R}, \mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R}\}.$$

Use pointwise operations:

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x).$$

For each $a \in \mathbb{R}$, the map

$$\text{ev}_a : \mathcal{F}(\mathbb{R}, \mathbb{R}) \rightarrow \mathbb{R}, \quad f \mapsto f(a).$$

is a ring homomorphism.

Example: Additive Isomorphism Need Not Preserve Multiplication

As abelian groups,

$$\mathbb{Z} \simeq 2\mathbb{Z}$$

by

$$\phi(x) = 2x.$$

But this is not a ring homomorphism:

$$\phi(xy) = 2xy, \quad \phi(x)\phi(y) = 4xy.$$

A ring homomorphism must preserve both addition and multiplication.

Part II

Ideals and Quotient Rings

Example: Historical Motivation for Ideals

Kummer introduced **ideal numbers** while studying Fermat's Last Theorem.

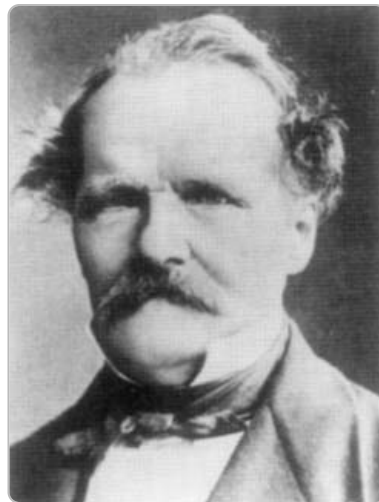
The obstruction was failure of unique factorization in some rings of algebraic integers.

Dedekind later replaced ideal numbers by **ideals**:

subsets that behave like generalized multiples.

For us today:

ideals are the right kernels for ring homomorphisms.



Ernst Eduard Kummer

Definition: Ideals

Let R be a commutative ring.

Definition. An **ideal** $I \subseteq R$ is a subset such that:

1. I is an additive subgroup of $(R, +)$;
2. for every $r \in R$ and every $a \in I$,

$$ra \in I.$$

If $I \neq R$, then I is a **proper ideal**.

Proposition: Kernels Are Ideals

Let $\varphi : R \rightarrow S$ be a ring homomorphism of commutative rings. Then

$$\ker \varphi = \{r \in R : \varphi(r) = 0\}$$

is an ideal of R .

Example:

$$\pi : \mathbb{Z} \rightarrow \mathbb{Z}_n, \quad a \mapsto \bar{a}, \quad \ker \pi = n\mathbb{Z}.$$

Proof: Kernels Are Ideals

If $a, b \in \ker \varphi$, then

$$\varphi(a - b) = \varphi(a) - \varphi(b) = 0.$$

So $\ker \varphi$ is an additive subgroup.

If $r \in R$ and $a \in \ker \varphi$, then

$$\varphi(ra) = \varphi(r)\varphi(a) = 0.$$

Thus $ra \in \ker \varphi$.

Definition: Principal Ideals

Let R be a commutative ring with unity and let $a \in R$.

The **principal ideal generated by a** is

$$(a) = Ra = \{ra : r \in R\}.$$

Examples:

$$(n) = n\mathbb{Z} \subset \mathbb{Z}, \quad (x - a) \subset F[x].$$

This notation also appears inside quotient rings such as $\mathbb{Z}_n$.

Whenever I is an ideal, we can form the quotient ring R/I .

Definition: Principal Ideal Domains

Let D be an integral domain.

A **principal ideal domain**, or **PID**, is an integral domain in which every ideal is principal.

That is, for every ideal $I \subseteq D$, there exists $a \in D$ such that

$$I = (a).$$

Examples:

$$\mathbb{Z}, \quad F[x].$$

A typical non-example is

$$F[x, y],$$

where the ideal (x, y) is not principal.

Definition: Quotient Rings

Let R be a commutative ring and let $I \subseteq R$ be an ideal.

The quotient ring R/I is the set of cosets

$$r + I = \{r + i : i \in I\}.$$

Addition and multiplication are defined by

$$(r + I) + (s + I) = (r + s) + I,$$

$$(r + I)(s + I) = rs + I.$$

The quotient map

$$\pi : R \rightarrow R/I, \quad r \mapsto r + I$$

is a ring homomorphism with $\ker \pi = I$.

Example: Quotient Rings

For $n \geq 1$,

$$\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}_n.$$

The coset $a + n\mathbb{Z}$ corresponds to $\bar{a}$.

In polynomial rings,

$$F[x]/(x - a) \simeq F$$

by evaluation at a .

Proposition: Ideals in $\mathbb{Z}_n$

Ideals of $\mathbb{Z}_n$ correspond to divisors of n .

For $d \mid n$,

$$(d) = d\mathbb{Z}_n = \{\bar{0}, \bar{d}, \overline{2d}, \dots\}.$$

Every ideal of $\mathbb{Z}_n$ has this form.

The larger d is, the smaller (d) is.

Example: Ideals of $\mathbb{Z}_{12}$

The divisors of 12 are

1, 2, 3, 4, 6, 12.

So the ideals are

(1), (2), (3), (4), (6), (12).

The proper ideals among them are

(2), (3), (4), (6), (12).

Definition: Maximal Ideals

Let R be a commutative ring with unity.

An ideal $\mathfrak{m} \subsetneq R$ is **maximal** if there is no ideal I with

$$\mathfrak{m} \subsetneq I \subsetneq R.$$

Key test:

$$\mathfrak{m} \text{ is maximal} \iff R/\mathfrak{m} \text{ is a field.}$$

Definition: Prime Ideals

Let R be a commutative ring with unity.

An ideal $\mathfrak{p} \subsetneq R$ is **prime** if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

Key test:

$$\mathfrak{p} \text{ is prime} \iff R/\mathfrak{p} \text{ is an integral domain.}$$

Always:

$$\text{maximal} \implies \text{prime}.$$

Example: Maximal Ideals in $\mathbb{Z}_{12}$

In $\mathbb{Z}_{12}$:

$$(2) \quad \text{and} \quad (3)$$

are maximal.

Indeed,

$$\mathbb{Z}_{12}/(2) \simeq \mathbb{Z}_2, \quad \mathbb{Z}_{12}/(3) \simeq \mathbb{Z}_3.$$

They are also prime.

For $d \mid n$:

$$(\mathbb{Z}_n)/(d) \simeq \mathbb{Z}_d.$$

Hence

$$(d) \subset \mathbb{Z}_n \text{ is maximal} \iff d \text{ is prime.}$$

Example: Prime Ideals in $\mathbb{Z}_{12}$

The same ideals

$$(2) \quad \text{and} \quad (3)$$

are prime, because their quotient rings are fields.

The ideal (4) is not prime:

$$\bar{2} \cdot \bar{2} = \bar{4} \in (4), \quad \bar{2} \notin (4).$$

The ideal $(12) = 0$ is not prime because $\mathbb{Z}_{12}$ has zero divisors.

Proposition: Ideals in $F[x]$

For $a \in F$,

$$(x - a) \subset F[x]$$

is maximal because

$$F[x]/(x - a) \simeq F.$$

More generally, if $p(x) \in F[x]$ is irreducible, then

$$(p(x))$$

is maximal.

Example: Prime But Not Maximal

In a commutative ring with unity:

$$\text{maximal} \implies \text{prime}.$$

But prime need not imply maximal.

Example:

$$(x) \subset F[x, y].$$

Since

$$F[x, y]/(x) \simeq F[y],$$

and $F[y]$ is an integral domain but not a field, (x) is prime but not maximal.

Example: Point Ideals in $F[x, y]$

For a point $(a, b) \in F^2$,

$$\mathfrak{m}_{(a,b)} = (x - a, y - b).$$

This is maximal because

$$F[x, y]/(x - a, y - b) \simeq F.$$

Geometric meaning:

$$(x - a, y - b) = \text{polynomials vanishing at } (a, b).$$

Definition: Radical of an Ideal

Let $I \subseteq R$ be an ideal in a commutative ring.

The **radical** of I is

$$\sqrt{I} = \{r \in R : r^n \in I \text{ for some } n \geq 1\}.$$

If $\sqrt{I} = I$, then I is a **radical ideal**.

The radical $\sqrt{I}$ is again an ideal.

Definition: Nilradical

The radical of the zero ideal is the **nilradical**:

$$\sqrt{(0)} = \{r \in R : r^n = 0 \text{ for some } n \geq 1\}.$$

It is the set of nilpotent elements of R .

Example: Radical Ideals

In $F[x]$:

$$\sqrt{((x-a)^m)} = (x-a).$$

In $F[x, y]$:

$$\sqrt{(x^2, y)} = (x, y).$$

Radical removes repeated or nilpotent behavior.

Example: Radical in $\mathbb{Z}_{12}$

In $\mathbb{Z}_{12}$,

$$\sqrt{(4)} = (2).$$

Reason:

$$\bar{2}^2 = \bar{4} \in (4),$$

so $\bar{2} \in \sqrt{(4)}$.

This is another way nilpotent-like behavior appears in quotient rings.

Part III

Polynomial Rings

Definition: Polynomial Rings

Let R be a commutative ring with unity.

The polynomial ring over R is

$$R[x] = \{a_0 + a_1x + \cdots + a_nx^n : a_i \in R\}.$$

The symbol x is an **indeterminate**, not an element of R .

The inclusion $R \rightarrow R[x]$ sends r to the constant polynomial r .

Definition: Formal Polynomial Viewpoint

A polynomial can be modeled as a formal sum

$$\sum_{i=0}^{\infty} a_i x^i$$

where all but finitely many a_i are zero.

This avoids ambiguity:

$$1 + x = 1 + x + 0x^2 = 1 + x + 0x^2 + 0x^3 + \cdots .$$

Definition: Addition in $R[x]$

Write polynomials as finite R -linear combinations of the monomials x^i :

$$f = \sum_{i \geq 0} a_i x^i, \quad g = \sum_{i \geq 0} b_i x^i,$$

where $a_i = b_i = 0$ for all but finitely many i .

Addition is defined coefficientwise:

$$f + g = \sum_{i \geq 0} (a_i + b_i) x^i.$$

Example in $\mathbb{Z}_5[x]$:

$$(3x^2 + 4x + 1) + (4x^2 + 2) = 2x^2 + 4x + 3.$$

Definition: Multiplication in $R[x]$

Let

$$f = \sum_{i=0}^m a_i x^i, \quad g = \sum_{j=0}^n b_j x^j.$$

The product is

$$fg = \sum_{k=0}^{m+n} c_k x^k,$$

where

$$c_k = \sum_{i+j=k} a_i b_j.$$

This is the usual distributive multiplication:

$$(a_i x^i)(b_j x^j) = a_i b_j x^{i+j}.$$

Example: Multiplication Depends on R

In $\mathbb{Z}[x]$:

$$(x + 1)(x^2 + 2) = x^3 + x^2 + 2x + 2.$$

In $\mathbb{Z}_2[x]$:

$$(x + 1)^2 = x^2 + 2x + 1 = x^2 + 1.$$

The coefficient ring matters.

Lemma: Degree over Domains

Let D be an integral domain.

For a nonzero polynomial

$$f(x) = a_n x^n + \cdots + a_1 x + a_0, \quad a_n \neq 0,$$

the **degree** is

$$\deg f = n.$$

If $f, g \in D[x]$ are nonzero, then

$$\deg(fg) = \deg f + \deg g.$$

Proposition: Polynomial Rings Preserve Domains

Let D be an integral domain.

Then $D[x]$ is an integral domain.

Indeed, if $0 \neq f, g \in D[x]$, then

$$\deg(fg) = \deg f + \deg g.$$

So $fg \neq 0$.

$D[x]$ has no zero divisors.

Proposition: Units in $F[x]$

Let F be a field.

The units are exactly the nonzero constants:

$$F[x]^{\times} = F^{\times}.$$

Indeed, if $f(x)g(x) = 1$, then $\deg f + \deg g = 0$.

In particular, $F[x]$ is not a field, because x has no inverse.

Counterexample: Zero Divisors in $R[x]$

If the coefficient ring has zero divisors, then the polynomial ring does too.

In $\mathbb{Z}_6$:

$$\bar{2} \cdot \bar{3} = \bar{0}.$$

So in $\mathbb{Z}_6[x]$:

$$(2x + 2)(3x + 3) = 0.$$

The good behavior of $F[x]$ comes from no zero divisors in F .

Definition: Multivariable Polynomial Rings

For a commutative ring with unity R ,

$$R[x_1, \dots, x_n]$$

is the ring of finite R -linear combinations of monomials

$$x_1^{a_1} \cdots x_n^{a_n}.$$

Here $a_1, \dots, a_n \in \mathbb{Z}_{\geq 0}$.

For example,

$$R[x, y] = (R[x])[y].$$

Part IV

Algebras, Free Algebras, and Evaluation

Definition: R -Modules

Let R be a commutative ring with unity.

An R -module is an abelian group M with scalar multiplication

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm,$$

such that

$$(r + s)m = rm + sm, \quad r(m + n) = rm + rn,$$

$$(rs)m = r(sm), \quad 1m = m.$$

Examples: vector spaces, R^n , and ideals $I \subseteq R$.

Definition: R -Module Homomorphisms

Let M, N be R -modules.

An **R -module homomorphism** is a map $\psi : M \rightarrow N$ such that

$$\psi(m_1 + m_2) = \psi(m_1) + \psi(m_2),$$

and

$$\psi(rm) = r\psi(m) \quad (r \in R, m \in M).$$

Definition: R -Algebras

Let R be a commutative ring with unity.

An R -algebra is a ring A with unity together with a unital ring homomorphism

$$R \rightarrow A$$

whose image lies in the center of A .

A **commutative R -algebra** is an R -algebra whose underlying ring is commutative.

Then A is an R -module by

$$r \cdot a = ra = ar.$$

Examples:

$$R, \quad M_n(F), \quad R[x], \quad R[x_1, \dots, x_n].$$

If $F \leq E$ is a field extension, then E is an F -algebra.

Definition: R -Algebra Homomorphisms

Let A and B be commutative R -algebras.

An **R -algebra homomorphism** is a map $\varphi : A \rightarrow B$ that is both

a unital ring homomorphism and an R -module homomorphism.

The R -module condition means

$$\varphi(ra) = r\varphi(a) \quad (r \in R, a \in A).$$

Since $\varphi(1_A) = 1_B$, this carries the scalar r in A to the scalar r in B .

Theorem: Universal Property of Polynomial Algebras

Let $R[x_s \mid s \in S]$ be the polynomial ring whose variables are indexed by S .

Let

$$i : S \rightarrow R[x_s \mid s \in S], \quad s \mapsto x_s.$$

For every commutative R -algebra A , precomposition with i gives a bijection

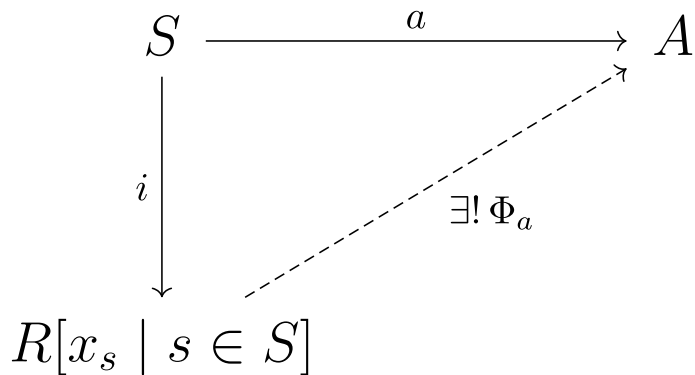
$$i^* : \operatorname{Hom}_{R\text{-alg}}(R[x_s \mid s \in S], A) \longrightarrow \operatorname{Map}(S, A).$$

Theorem: Universal Property Diagram

Equivalently, each map $a : S \rightarrow A$ determines a unique R -algebra homomorphism Φ_a .

The defining condition is

$$\Phi_a(x_s) = a(s) \quad (s \in S).$$



Corollary: Polynomial Algebras Are Free

The universal property says:

every set map $a : S \rightarrow A$ extends uniquely to an R -algebra homomorphism

$$\Phi_a : R[x_s \mid s \in S] \rightarrow A.$$

This is what it means to say:

$$R[x_s \mid s \in S]$$

is the **free commutative R -algebra on the set S** .

In particular, $R[x]$ is free on one generator, and $R[x_1, \dots, x_n]$ is free on n generators.

Proof: Polynomial Rings Are Free

Given a set map $a : S \rightarrow A$, define Φ_a by

$$\Phi_a(x_s) = a(s), \quad \Phi_a(r) = r.$$

On a monomial,

$$\Phi_a(r x_{s_1}^{e_1} \cdots x_{s_m}^{e_m}) = r a(s_1)^{e_1} \cdots a(s_m)^{e_m}.$$

Extend by finite sums. This gives an R -algebra homomorphism because multiplication of monomials corresponds to adding exponents.

Uniqueness: every polynomial is built from scalars, addition, multiplication, and the generators x_s .

Definition: Evaluation Homomorphism

Let A be a commutative R -algebra and let $a : S \rightarrow A$ be a set map.

The unique homomorphism

$$\text{ev}_a : R[x_s \mid s \in S] \rightarrow A, \quad x_s \mapsto a(s)$$

is called the **evaluation homomorphism** at a .

If $S = \{1, \dots, n\}$, then $a : S \rightarrow A$ is the same data as a tuple

$$(a_1, \dots, a_n) \in A^n,$$

and

$$\text{ev}_a : R[x_1, \dots, x_n] \rightarrow A, \quad x_i \mapsto a_i.$$

Definition: Polynomial Functions from Evaluation

Let A be a commutative R -algebra.

For $f \in R[x_1, \dots, x_n]$, the **polynomial function defined by f on A** is

$$f_A : A^n \rightarrow A, \quad (a_1, \dots, a_n) \mapsto \text{ev}_{(a_1, \dots, a_n)}(f).$$

In one variable, if $f = r_0 + r_1x + \dots + r_dx^d$, then

$$\text{ev}_a(f) = r_0 + r_1a + \dots + r_da^d.$$

Example: A Nonzero Polynomial Giving the Zero Function

Let $F = \mathbb{F}_p$ and let

$$f(x) = x^p - x \in F[x].$$

For every $a \in F$, Fermat's theorem gives

$$f_F(a) = a^p - a = 0.$$

Thus $f_F : F \rightarrow F$ is the zero function, but $f \neq 0$ as an element of $F[x]$.

Proposition: Evaluation Preserves Operations

Let A be a commutative R -algebra and let $a : S \rightarrow A$.

For $f, g \in R[x_s \mid s \in S]$,

$$\text{ev}_a(f + g) = \text{ev}_a(f) + \text{ev}_a(g),$$

$$\text{ev}_a(fg) = \text{ev}_a(f) \text{ev}_a(g).$$

This is not an extra calculation: it is exactly the statement that ev_a is an R -algebra homomorphism.

Example: A Point Ideal in Two Variables

Skip if short on time.

Take $R = A = \mathbb{Q}$ and the point $(2, 3) \in \mathbb{Q}^2$.

The evaluation map is

$$\text{ev}_{(2,3)} : F[x, y] \rightarrow F, \quad x \mapsto 2, \quad y \mapsto 3.$$

Its kernel is the point ideal

$$(x - 2, y - 3).$$

This is the multivariable analogue of $\ker(\text{ev}_a) = (x - a)$.

Summary

Today we built the ring-theoretic language needed for polynomials.

- rings, units, zero divisors, integral domains;
- ideals, quotient rings, prime and maximal ideals;
- polynomial rings and polynomial evaluation;
- polynomial algebras as free commutative algebras.

Looking Ahead

Next we use polynomial division and localization to study roots.

The main bridge is

$$D \hookrightarrow \operatorname{Frac}(D).$$